

RoadSense AI Backend Integration Guide

Current Status

- YOLO model deployed on AWS EC2.
- FastAPI exposes GET / and POST /predict.
- Model: models/best.pt.

Architecture

Flutter → Fastify Backend → EC2(FastAPI+YOLO) → JSON → Backend → PostgreSQL(RDS)+S3 → Flutter.

Backend Responsibilities

1. Receive image/video.
2. Upload original media to S3.
3. Send image to EC2 /predict.
4. Receive detections, severity and score.
5. Store report (S3 URL, AI JSON, GPS, user, timestamps).
6. Return report to Flutter.

Report Model

reportId, userId, mediaUrl, latitude, longitude(PostGIS), aiResponse(JSON), highestSeverity, damageScore, status, createdAt.

Demo Checklist

- Connect Fastify with EC2 API.
- Connect S3 upload.
- Save report in PostgreSQL.
- Fetch reports in Flutter.
- Draw AI bounding boxes on the displayed S3 image.

Future

Docker, Nginx, HTTPS, better dataset, retraining, production deployment.